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14362 — High Performance Computing for Big Data

Section 1

# Predicting Clinical Significance of Human Genetic Variants

A Big Data Pipeline using MongoDB, Apache Spark, and MLlib

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## Table of Contents

|                                                  |    |
|--------------------------------------------------|----|
| 1. Introduction.....                             | 3  |
| 2. Dataset.....                                  | 3  |
| 3. System Architecture.....                      | 4  |
| 4. Phase 1: MongoDB Ingestion & Exploration..... | 5  |
| 4.1 Data Ingestion.....                          | 5  |
| 4.2 Exploratory Queries.....                     | 5  |
| 5. Phase 2: PySpark ETL & Analysis.....          | 7  |
| 5.1 Loading Data into Spark.....                 | 7  |
| 5.2 Data Cleaning & Transformations.....         | 7  |
| 5.3 Feature Engineering.....                     | 8  |
| 5.4 Dataset Joins.....                           | 8  |
| 5.5 SparkSQL Analytical Results.....             | 9  |
| 6. Phase 3: Machine Learning with MLlib.....     | 12 |
| 6.1 Feature Preparation.....                     | 12 |
| 6.2 Model: Random Forest Classifier.....         | 12 |
| 6.3 Hyperparameter Tuning (CrossValidator).....  | 13 |
| 6.4 Model Evaluation.....                        | 13 |
| 6.5 Feature Importance.....                      | 14 |
| 7. Discussion.....                               | 14 |
| 8. Conclusion.....                               | 15 |

## 1. Introduction

Genetic variant classification is a fundamental challenge in clinical genomics. The ClinVar database, maintained by the National Center for Biotechnology Information (NCBI), aggregates millions of variant-level submissions from clinical laboratories, research institutions, and expert panels worldwide. Each variant is assessed for its clinical significance — most critically, whether it is Pathogenic, Benign, or of Uncertain Significance (VUS).

This project implements a full end-to-end big data pipeline to address a practical bioinformatics problem: given a variant's metadata — its gene, type, chromosome position, review confidence, and submission history — can we automatically predict whether it will be classified as Pathogenic or Benign? This mirrors real clinical workflows where labs routinely encounter novel variants that require rapid triage.

The pipeline is implemented across three phases: document-oriented ingestion and exploration in MongoDB, distributed ETL and analytical querying in Apache Spark, and predictive modeling using Spark MLlib's Random Forest classifier with distributed cross-validation.

## 2. Dataset

All data is sourced from the NCBI ClinVar FTP repository ([https://ftp.ncbi.nlm.nih.gov/pub/clinvar/tab\\_delimited/](https://ftp.ncbi.nlm.nih.gov/pub/clinvar/tab_delimited/)). Three files were used across the pipeline:

| File                      | Size    | Records    | Description                                                                                   |
|---------------------------|---------|------------|-----------------------------------------------------------------------------------------------|
| variant_summary.txt.gz    | ~420 MB | 8,978,110  | One row per variant-assembly pair. Primary data source for all three phases.                  |
| submission_summary.txt.gz | ~230 MB | 6.3M lines | Per-submission details. Join key: VariationID. Used to compute submission counts per variant. |

The dataset was filtered to the GRCh37 genome assembly to avoid duplicating variants that appear in multiple assemblies, and further restricted to variants with unambiguous Pathogenic or Benign labels for the machine learning phase.

## 3. System Architecture

The entire pipeline runs inside a containerized environment managed by Docker Compose. No local installation of MongoDB, Spark, or Python dependencies is required beyond Docker Desktop. The four services are:

| Service | Image | Port | Role |
|---------|-------|------|------|
|---------|-------|------|------|

| MongoDB      | mongo:7.0                            | 27017       | Document store for raw variant data. Phase 1 target. |
|--------------|--------------------------------------|-------------|------------------------------------------------------|
| Jupyter Lab  | jupyter/pyspark-notebook:spark-3.5.0 | 8888        | Notebook environment. Runs all three phases.         |
| Spark Master | apache/spark:3.5.0                   | 8080 / 7077 | Cluster coordinator. Distributes Spark jobs.         |
| Spark Worker | apache/spark:3.5.0                   | 8081        | 4 cores, 4 GB memory. Executes distributed tasks.    |

MongoDB is accessed from Jupyter via the service hostname (mongodb:27017) for PyMongo operations. Spark reads MongoDB through the host bridge IP (172.20.0.1) because Spark executors operate outside the Compose network context when submitting to a standalone cluster. The MongoDB Spark Connector (org.mongodb.spark:mongo-spark-connector\_2.12:10.4.0) bridges the two systems.

## 4. Phase 1: MongoDB Ingestion & Exploration

### 4.1 Data Ingestion

The variant\_summary.txt.gz file was streamed directly from disk into MongoDB using Python's built-in csv and gzip modules — no pandas was used at any point. Documents were inserted in batches of 5,000 for performance, avoiding single-document round-trips across 8.9 million records.

```
with gzip.open("variant_summary.txt.gz", mode="rt", encoding="utf-8") as f:
    reader = csv.DictReader(f, delimiter="\t")
    for row in reader:
        batch.append(row)
        if len(batch) >= 5000:
            collection.insert_many(batch)
            batch = []
```

Total documents ingested: 8,978,110

### 4.2 Exploratory Queries

Three PyMongo aggregation pipelines were run directly against the document store to understand the dataset before any transformation.

Total Variant Count

8,978,110 total variants across all chromosomes and assemblies.

Distribution of Variant Types

| Variant Type | Count | Share (%) |
|--------------|-------|-----------|
|--------------|-------|-----------|

|                           |           |       |
|---------------------------|-----------|-------|
| Single Nucleotide Variant | 8,271,960 | 92.1% |
| Deletion                  | 346,203   | 3.9%  |
| Duplication               | 148,541   | 1.7%  |
| Microsatellite            | 77,426    | 0.9%  |
| Insertion                 | 28,397    | 0.3%  |
| Copy Number Loss          | 30,136    | 0.3%  |
| Copy Number Gain          | 32,514    | 0.4%  |
| Indel                     | 38,121    | 0.4%  |
| Other                     | 4,812     | 0.05% |

Single nucleotide variants (SNVs) dominate the dataset at over 92%, reflecting the clinical focus on point mutations.

#### Variants per Chromosome

Chromosome 1 carries the highest variant load (808,005), consistent with it being the longest human chromosome. The mitochondrial genome (MT) contains 6,248 variants, and 9,360 variants are unassigned (na) — typically structural variants not tied to a specific chromosomal locus. The sex chromosomes (X: 389,166; Y: 7,022) show the expected asymmetry due to differential coverage in clinical databases.

## 5. Phase 2: PySpark ETL & Analysis

### 5.1 Loading Data into Spark

The MongoDB collection was read into a Spark DataFrame using the MongoDB Spark Connector. The Spark session was configured to connect to the standalone cluster at `spark://spark-master:7077` with 1 GB driver memory and 3 GB executor memory.

```
spark = SparkSession.builder\
    .appName("BigDataSpark")\
    .master("spark://spark-master:7077")\
    .config("spark.jars.packages", "org.mongodb.spark:mongo-spark-connector_2.12:10.4.0")\
    .getOrCreate()
```

### 5.2 Data Cleaning & Transformations

The following cleaning steps were applied in sequence:

- Replaced dash placeholders ("-") with null values across all string columns.
- Cast key columns to appropriate types: AlleleID → Int, Start/Stop → Int, NumberSubmitters → Int.
- Filtered to a single genome assembly (GRCh37) to eliminate duplicate variant representations.
- Dropped rows where Chromosome, Start, or Stop were null (~29 rows removed).

- Stripped whitespace from ClinicalSignificance and mapped compound labels to four simplified categories: Pathogenic, Benign, Uncertain, and Other.

### 5.3 Feature Engineering

Two derived columns were computed from existing fields:

| Column         | Formula                               | Purpose                                      |
|----------------|---------------------------------------|----------------------------------------------|
| variant_length | Stop – Start + 1                      | Structural size of the variant in base pairs |
| review_score   | Ordinal mapping of ReviewStatus (0-4) | Numeric confidence score for ML              |
| is_pathogenic  | 1 if Pathogenic, 0 if Benign          | Binary label for classification              |

The review\_score mapping assigns: 0 = no criteria, 1 = single submitter, 2 = multiple submitters no conflicts, 3 = expert panel, 4 = practice guideline. Variants labeled Uncertain Significance or with ambiguous classifications were excluded from the machine learning phase to maintain a clean binary target.

### 5.4 Dataset Joins

Two external datasets were joined to enrich each variant record:

- **Gene-level aggregates:** Variants were grouped by GeneSymbol to compute total\_variants and pathogenic\_variants per gene. These aggregates were joined back onto the main DataFrame to give each variant context about its gene's overall pathogenicity profile.
- **Submission summary:** The submission\_summary.txt.gz file was read directly into Spark with a defined schema. Submissions were aggregated per VariationID to produce sub\_num\_submissions and sub\_num\_submitters, then joined onto the main DataFrame.

### 5.5 SparkSQL Analytical Results

Top 10 Genes with Most Pathogenic Variants

| Rank | Gene Symbol | Pathogenic Variants |
|------|-------------|---------------------|
| 1    | BRCA2       | 5,375               |
| 2    | NF1         | 5,025               |
| 3    | BRCA1       | 4,186               |
| 4    | ATM         | 3,309               |
| 5    | DMD         | 2,790               |
| 6    | FBN1        | 2,566               |
| 7    | APC         | 2,401               |
| 8    | MSH6        | 2,090               |

|    |      |       |
|----|------|-------|
| 9  | MSH2 | 2,036 |
| 10 | PKD1 | 1,692 |

BRCA2 and BRCA1 — both established breast and ovarian cancer susceptibility genes — rank in the top 3. NF1 (neurofibromatosis type 1) ranks second, reflecting its large gene size and high mutation rate. The mismatch repair genes MSH6 and MSH2 appear in positions 8 and 9, consistent with their role in Lynch syndrome.

#### Variant Type Distribution: Pathogenic vs. Benign

| Variant Type              | Benign Count | Pathogenic Count |
|---------------------------|--------------|------------------|
| Single nucleotide variant | 241,981      | 111,604          |
| Deletion                  | 12,983       | 80,561           |
| Duplication               | 10,005       | 28,468           |
| Microsatellite            | 6,137        | 8,574            |
| Copy number loss          | 1,345        | 7,977            |
| Indel                     | 330          | 4,621            |
| Insertion                 | 1,946        | 4,492            |
| Copy number gain          | 2,149        | 3,123            |

While SNVs dominate in absolute count, structural variants (deletions, duplications, copy number changes) show a markedly higher pathogenic-to-benign ratio. Deletions have a 6.2:1 pathogenic-to-benign ratio versus 0.46:1 for SNVs, suggesting that large structural changes are disproportionately damaging.

#### Conflict Resolution Rates by Gene (Pathogenic Ratio)

Genes were ranked by the proportion of pathogenic variants among all classified variants. Top results (minimum 50 total variants shown):

| Gene Symbol | Pathogenic Variants | Total Variants | Pathogenic Ratio |
|-------------|---------------------|----------------|------------------|
| GLA         | 817                 | 863            | 94.7%            |
| NF1         | 5,025               | 5,325          | 94.4%            |
| TYR         | 207                 | 222            | 93.2%            |
| RMRP        | 201                 | 216            | 93.1%            |
| GBA1        | 137                 | 147            | 93.2%            |

GLA (Fabry disease gene) and NF1 show the highest pathogenic ratios among high-volume genes, indicating that most reported variants in these genes are clinically significant.

### Diseases with Highest Pathogenic Variant Proportion

Diseases were ranked by the proportion of their associated variants classified as pathogenic (minimum 10 variants). Several diseases show 100% pathogenic classification:

| Disease                        | Pathogenic | Total | Proportion |
|--------------------------------|------------|-------|------------|
| ABCA4-related retinopathy      | 83         | 83    | 100%       |
| Embryonal rhabdomyosarcoma     | 68         | 68    | 100%       |
| Homocystinuria                 | 61         | 61    | 100%       |
| CHD7-related CHARGE syndrome   | 43         | 43    | 100%       |
| Morquio syndrome               | 43         | 43    | 100%       |
| Erythrocytosis, familial, 6    | 38         | 38    | 100%       |
| Crigler-Najjar syndrome type 1 | 17         | 17    | 100%       |

A 100% pathogenic proportion indicates that every recorded submission for variants associated with these diseases has been classified as pathogenic — consistent with rare Mendelian disorders where the gene-disease relationship is well-established and variants are typically causative.

## 6. Phase 3: Machine Learning with MLlib

The goal of Phase 3 is to train a binary classifier that predicts whether a variant is Pathogenic (label = 1) or Benign (label = 0) based on its metadata features. All computation runs on the distributed Spark cluster — no scikit-learn or local ML libraries were used.

### 6.1 Feature Preparation

The following features were selected, cast to Double, and null-filled before model training:

| Feature             | Type    | Description                                 |
|---------------------|---------|---------------------------------------------|
| review_score        | Numeric | Ordinal review confidence (0–4)             |
| variant_length      | Numeric | Stop – Start + 1 (base pairs)               |
| NumberSubmitters    | Numeric | Submitters from variant_summary             |
| sub_num_submissions | Numeric | Total submissions from submission_summary   |
| sub_num_submitters  | Numeric | Distinct submitters from submission_summary |
| total_variants      | Numeric | Total variants in the same gene             |
| pathogenic_variants | Numeric | Pathogenic variant count in the             |

|            |             |                                          |
|------------|-------------|------------------------------------------|
|            |             | same gene                                |
| Type       | Categorical | Variant type (StringIndexer encoded)     |
| Chromosome | Categorical | Chromosome label (StringIndexer encoded) |

Categorical features were encoded using StringIndexer with handleInvalid="keep" to gracefully handle unseen categories at inference time. All nine features were assembled into a single feature vector using VectorAssembler.

## 6.2 Model: Random Forest Classifier

A Random Forest Classifier was selected for its robustness to outliers, ability to handle mixed feature types, and built-in feature importance scores. It also parallelizes naturally across the Spark cluster — each tree is built independently.

The dataset was split 80% training / 20% test with a fixed seed for reproducibility. The training set consisted of approximately 421,497 variants after filtering to Pathogenic and Benign labels only.

## 6.3 Hyperparameter Tuning (CrossValidator)

A CrossValidator with 3-fold cross-validation was used to evaluate all combinations of:

| Parameter | Values Tested | Description                   |
|-----------|---------------|-------------------------------|
| maxDepth  | 5, 10         | Maximum depth of each tree    |
| numTrees  | 20, 50        | Number of trees in the forest |

This produced 4 parameter combinations × 3 folds = 12 full model fits on the Spark cluster. The combination with the highest AUC-ROC on the validation fold was selected as the best model.

## 6.4 Model Evaluation

The best model was evaluated on the held-out test set using the following metrics:

| Metric    | Value  |
|-----------|--------|
| AUC-ROC   | 0.8981 |
| Accuracy  | 80.71% |
| Precision | 80.93% |
| Recall    | 80.60% |
| F1 Score  | 80.64% |

Confusion Matrix:

|                   | Predicted Benign | Predicted Pathogenic |
|-------------------|------------------|----------------------|
| Actual Benign     | 47,419           | 8,120                |
| Actual Pathogenic | 12,227           | 37,715               |

## 6.5 Feature Importance

After training, the Random Forest model produces a feature importance score for each input feature, representing the average reduction in impurity (Gini impurity) weighted by the fraction of samples reaching that split. The results indicate which features are most predictive of pathogenicity:

| Rank | Feature             | Importance Score |
|------|---------------------|------------------|
| 1    | pathogenic_variants | 0.3480           |
| 2    | Type_idx            | 0.3316           |
| 3    | review_score        | 0.0975           |
| 4    | variant_length      | 0.0924           |
| 5    | total_variants      | 0.0772           |

## 7. Discussion

The Phase 1 and Phase 2 analytical results reveal several patterns consistent with established clinical genomics knowledge. The dominance of SNVs in the dataset (92.1%) reflects the clinical community's focus on point mutations, yet structural variants show a far higher pathogenic-to-benign ratio — deletions at 6.2:1 versus 0.46:1 for SNVs. This suggests that while SNVs are the most common variant type, large structural alterations are disproportionately harmful, a well-known phenomenon in copy number variation research.

The gene-level analysis confirms that genes involved in well-characterized cancer syndromes (BRCA1/2, MSH2/6, APC) and neurodevelopmental conditions (NF1) accumulate the most pathogenic calls, as these are the most frequently sequenced genes in clinical practice. The high pathogenic ratio in GLA (Fabry disease) and RMRP (cartilage-hair hypoplasia) reflects the narrow variant spectrum in single-gene Mendelian disorders.

The review\_score feature, derived from ReviewStatus, captures the epistemic confidence assigned by ClinVar curators. It is expected to be among the most predictive features for the classifier since higher review confidence correlates with more established pathogenicity calls.

One methodological consideration: the gene-level pathogenic counts (pathogenic\_variants, total\_variants) were computed from the full dataset before the train/test split, introducing a mild form of data leakage. In a production system, these would be computed from training data only. For the purposes of this academic exercise, this approach follows the project specification.

## 8. Conclusion

This project demonstrates a complete big data pipeline for clinical variant classification at scale. Starting from 8.97 million raw ClinVar records, the pipeline performs document-oriented ingestion into MongoDB, distributed ETL and analytical querying in Apache Spark, and predictive modeling using a Random Forest classifier with distributed hyperparameter tuning.

The MongoDB exploration phase confirms that single nucleotide variants dominate the ClinVar database, while Chromosome 1 carries the highest variant burden. The Spark analysis reveals that structural variants — though less frequent — are disproportionately pathogenic, and that well-studied cancer and Mendelian disease genes accumulate the largest share of pathogenic calls.

The machine learning phase uses nine features extracted from variant metadata to train a Random Forest classifier, tuned via 3-fold cross-validation over four hyperparameter combinations. The containerized architecture ensures that the pipeline is fully reproducible and runnable on any machine with Docker Desktop installed, with no additional configuration required.

Future work could address the mild data leakage in gene-level feature computation, explore additional features such as population frequency (gnomAD), and extend the classification to handle Variants of Uncertain Significance (VUS) through a multi-class formulation.